Status	Finished
Started	Tuesday, 20 October 2026, 10:56 AM
Completed	Tuesday, 20 October 2026, 11:00 AM
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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It only works on small datasets
- c. It avoids the need for recursion
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **2**

Complete

Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Being the most efficient sorting algorithm
- b. Using the divide-and-conquer approach
- c. Bubbling up the largest element to its correct position with each pass
- d. Sorting data in a non-sequential manner

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **3**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. Searching
- b. Duplicates
- c. Selection
- d. All of the above

The correct answer is: All of the above

Question **4**

Complete

Mark 1.00 out of 1.00

What is a significant characteristic of Bubble Sort?

- a. It recursively sorts subproblems
- b. It divides the list into sublists
- c. It bubbles up the largest element in each pass
- d. It uses the heap data structure

The correct answer is: It bubbles up the largest element in each pass

Question **5**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Heap sort
- b. Selection sort
- c. Merge sort
- d. Quick sort

The correct answer is:
Merge sort

Question **6**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Insertion Sort
- b. Quick Sort
- c. Bubble Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **7**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Searching
- b. Sorting
- c. Rearranging
- d. Merging

The correct answer is: Sorting

Question **8**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Searching for data in a list
- b. Deleting data from a list
- c. Arranging data in a particular format
- d. Inserting data into a list

The correct answer is: Arranging data in a particular format

Question **9**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It is less efficient than custom sorting algorithms
- b. It only works with integer arrays
- c. It requires external libraries
- d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **10**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Finding a specific element in a list
- b. Arranging data in a specific order
- c. Removing duplicates from a list
- d. Merging two datasets

The correct answer is: Arranging data in a specific order

Question **11**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By sorting data in a single pass
- b. By breaking the input into smaller parts and merging them
- c. By using the bubble-up method
- d. By comparing elements sequentially

The correct answer is: By breaking the input into smaller parts and merging them

Question **12**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It makes the search operation slower
- b. It has no effect on the search operation
- c. It increases the number of comparisons needed
- d. It allows for faster searching

The correct answer is: It allows for faster searching

Question **13**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Quick Sort
- c. Bubble Sort
- d. Insertion Sort

The correct answer is: Bubble Sort

Question **14**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It divides the list into two halves, sorts each half, and then merges them
- b. It builds a sorted array one element at a time
- c. It repeatedly finds the minimum element and moves it to the sorted part of the list
- d. It compares adjacent elements and swaps them if necessary

The correct answer is: It divides the list into two halves, sorts each half, and then merges them

Question **15**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are split again into smaller sublists
- c. They are compared element by element
- d. They are discarded

The correct answer is: They are combined to form a single sorted list